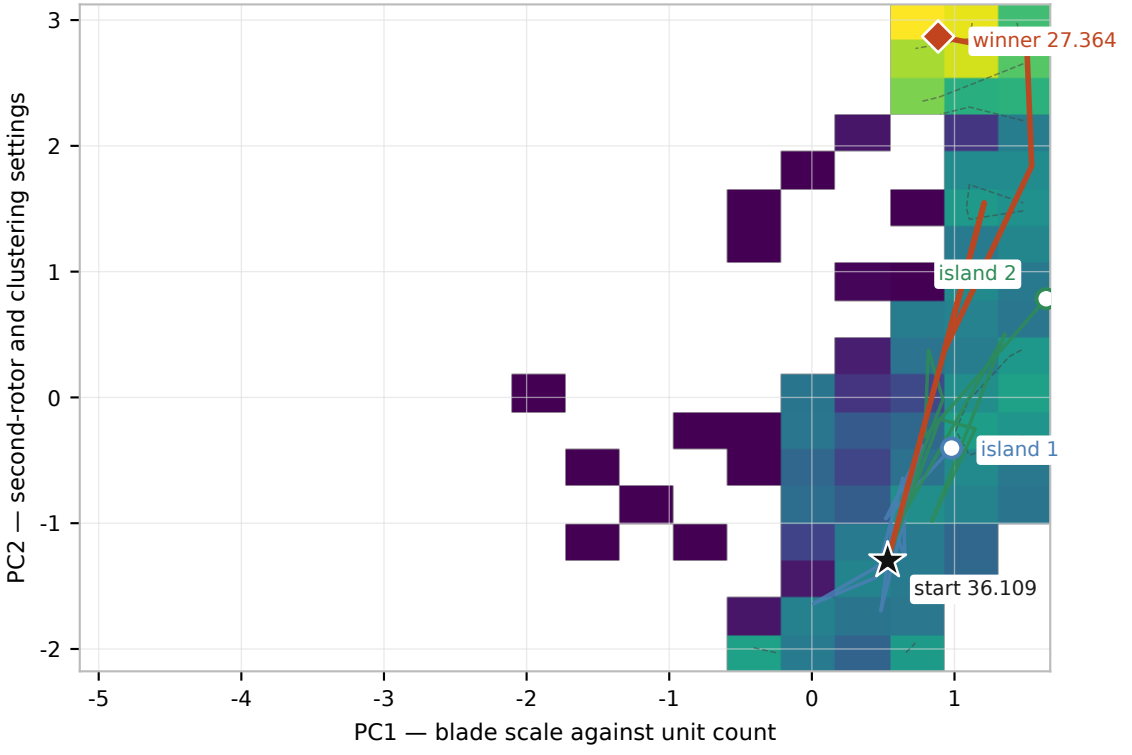
V13 campaign — parameter-space landscape

the 930-design screen · three DE islands · PC1-PC2 projection of the ten-gene search · colour is the search objective (kg)



cell mean of the search objective (kg)

bright = lower objective

feasible designs only;

blank cells carry none

PC1 — 22.4% of variance

+ PC1 ↑

larger main-rotor blades (x9)
wider spacing (x3), ground ring (x2)

− PC1 ↓

more TRPT lines (x4)
more rotors (x6)

direction reads from the loadings

THE THREE JOURNEYS

930 designs · three islands · 31 generations each

one common seed at 36.109 (kg)

winner: island 3 at 27.364 (kg)

each line: the island best-so-far path, one step per generation

PC2 — 17.3% of variance

+ PC2 ↑ deeper second-rotor bank (x8), smaller second blades (x10), rings biased toward the ground (x5)

− PC2 ↓ even rings, shallow second bank, larger second blades

source: island telemetry · data commit dd3cc6a · run a76c5e9 · era post-95c385d_wind-authority · rows NR-001, NR-005, NR-006, NR-044, NR-045

projection: z-scored 10-gene matrix, SVD · variance shares NR-044 · script commit e23f226